Technology Requirements

1. Frontend Technologies

1.1 Next.js: Next.js is a React based framework supporting server side rendering, static generation, and client side rendering. It is maintained by Vercel, AEGIS's deployment platform.

- **Justification:** Server side rendering improves initial load performance, supporting **NFR1.1.2** (minimal lag for concurrent candidate access). Its file based routing and modular component architecture support **NFR5.1.1**. Tight Vercel integration eliminates deployment setup overhead.
    - **Role in AEGIS:** Serves as the presentation layer for recruiter question management (**R1.1–R1.4**), adversarial generation preview and verification (**R2.3, R3.4**), candidate assessment execution (**R4.1–R4.3**) and assessment creation (**R5.1–R5.4**).

1.2 TailwindCSS: A utility first CSS framework that enables inline styling using predefined utility classes instead of custom CSS files.

- **Justification:** Predefined classes simplify building an accessible design (**NFR2.2.1**) with proper contrast ratios and keyboard navigation while preventing dead CSS accumulation (**NFR5.2.1**).
- **Role in AEGIS:** Styles all pages across recruiter and candidate interfaces (**R1.1–R5.4**).

2. Backend Technologies

2.1 FastAPI: A high performance Python web framework for REST APIs using type hints for automatic Pydantic data validation and Swagger UI documentation.

- **Justification:** Asynchronous execution handles concurrent requests without blocking (**NFR1.1.1, NFR1.1.2**) during candidate submissions and LLM/execution calls. Dependency injection allows mocking database and auth dependencies during unit testing (**NFR5.2.1**). Automatic Swagger UI documentation ensures endpoints are self documenting (**NFR5.1.1**).
    - **Role in AEGIS:** Acts as the application layer, processing frontend requests, orchestrating Gemini (**R2.2, R3.3**), Piston CE (**R4.4**) calls and executing scoring logic (**R4.5, R5.4.2**).

2.2 SQLAlchemy: A Python SQL toolkit and Object Relational Mapper (ORM) that maps database tables to Python classes.

- **Justification:** Decouples data access logic from route handlers (**NFR5.1.1**), ensuring database changes only require updating connection strings.By using selectinload, the system loads connected database tables together all at once, avoiding slow and repeated queries (**NFR1.1.2**).

- **Role in AEGIS:** Manages queries across all database tables.

2.3 Pydantic: A Python data validation library using type annotations to parse and validate runtime payloads.

- **Justification:** Enforces strict validation at API boundaries, returning structured error responses for malformed requests (**NFR2.1.1**).
- **Role in AEGIS:** Validates request/response schemas for question management (**R1.1**), adversarial generation (**R2.2**), verification decisions (**R3.4**), code submissions (**R4.4**), and assessment assignment (**R5.4**).

2.4 Supabase PostgreSQL: A hosted relational database providing a built in Studio UI, Row Level Security (RLS), and a standard PostgreSQL connection string.

- **Justification:** Free tier provides 500 MB of storage for proof of concept development. Standard PostgreSQL compatibility avoids vendor lock in. Built in RLS restricts candidate access solely to assigned assessments (**NFR3.1.1**), while the Studio UI reduces development overhead (**NFR5.2.1**).
  - **Role in AEGIS:** Serves as the persistent data store for question repositories (**R1.5.1–R1.5.3**), verification records (**R3.5**), candidate submissions and scores (**R4.2.1, R4.5.1**) and assessment assignments (**R5.4.2**).

3. AI and Other Technologies

3.1 Google Gemini API: Provides access to Google's Gemini 2.0 Flash model, featuring a 1 000 000 token context window and strong instruction following capabilities.

- **Justification:** Free tier allows 1,500 requests daily within budget constraints. The large context window accepts full prompt templates without truncation, and sufficient instruction following reliably embeds adversarial traps. Fast response times support system performance (**NFR1.1.2**).
  - **Role in AEGIS:** Generates adversarial variants from source questions (**R2.2.2–R2.2.3, R2.3.1**) and creates reference answers for verification (**R3.3.1, R3.4.2**).

3.2 Piston CE: An open source, self hosted code execution engine running code inside isolated Docker containers.

- **Justification:** Open source deployment incurs zero per request costs or rate limits (**NFR1.1.2**). Docker container isolation prevents candidate submissions from accessing the host file system or network (**NFR3.1.1**).
  - **Role in AEGIS:** Executes candidate Python submissions against

stored test cases (**R4.4.1–R4.4.2**) and returns output for scoring (**R4.5.1**).